

SANJAY M

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Executive Summary

B.E graduate in Electronics and Communication Engineering with an energetic approach towards learning. Automotive electronics enthusiast in Embedded systems with AI integration. A certified ECU tuner and Remapping. I enhance traditional embedded systems with intelligent solutions and am committed to innovation and continuous learning to drive growth and technical excellence.

Core Competencies

- **Programming Languages:** Embedded C | Advanced C++ | Assembly language | HTML/ CSS
- **EDA Tools:** Dev C++ | Docker Desktop | VS Code | Matlab | ReynICE Tuner Studio | Arduino IDE | Canva
- **Micro-controller:** Arduino Uno / Nano| ECU | ESP32 | Raspberry Pi | STM32 Nucleo F446RE
- **Protocols:** SPI | UART | CAN | Bluetooth BLE

Academic Projects

Ambient lighting effect using 555 timer • Calling bell system equipped with the light system. • SOS Light blinking system when the bell depressed.	<i>July 2024 - Nov 2024</i>
Six-Channel Transmitter for Drone Remote control • Standard 6 channel transceiver of range 1KM using Arduino Nano • Better efficiency covered by the NRF 24I04 module. • Operated at the frequency of 2.4 GHz ISM band for compatibility.	<i>Jan 2025 - May 2025</i>
Highway assist based on image text to speech conversion • Image captured by the camera is synthesized to speech via Matlab. • Time complexity of the system is less than 2 seconds.	<i>July 2025 - Nov 2025</i>
ECU simulation and diagnostic monitoring system using Docker • Whole system is containerized for the maximum compatibility. • System senses the temperature of the engine, speed of the vehicle. • LED light intensity varies based on the speed of the vehicle.	<i>Jan 2026 - May 2026</i>
ECU based smart anti-Glare headlight controller for Retrofitting • Reduces the intensity based on 2 modes of driving viz. City & Highway. • Direct control over headlamp for any kind of aftermarket kits. • Plug 'n' Play device for extended compatibility.	<i>July 2026 - Till date</i>

Activities

- Participated in the paper presentation on "**Handling of Digital Twin Technology**" at Tek Cluster 24, Kongunadu College of Engineering and Technology, Trichy, TN. **Sept 21, 2024**
- Presented a paper on "**Digital Twin Technology in Automobile**" at Francis Xavier Engineering College, Tirunelveli, TN. **March 21, 2025**
- Participated in the "**Unmanned Aerial Vehicle (UAV) Communication**" seminar organized by IETE Students' Forum and TiHAN by IIT Hyderabad. **Jan 30, 2026**
- Showcased a **Swarm-intelligence-based adaptive path planning algorithm for Unmanned Aerial Vehicles (UAVs)** at the 10th National Techno Exhibition Project Contest, Dr. Ambedkar Institute of Technology, Bangalore, KA. **April 13, 2026**
- Attended the "**Microcontroller Workshop with ATmega32**" by NPTEL+Workshop at IIT Kharagpur, West Bengal. **June 8 - 13, 2026**

Education

Mepco Schlenk Engineering College, Sivakasi. B.E in Electronics and Communication Engineering	2023 - Till date Latest GPA: 7.894 CGPA: 7.07
Glazebrooke Public School, Salem. HSC (CBSE)	2023 76.2%
Glazebrooke Public School, Salem. SSLC (CBSE)	2021 85.6%

Certification & Achievements

- ECU Tuning and Remapping - **Techmaghi**
- Digital SOC Design Verification (**VAC**)
- NPTEL Enthusiast - **NPTEL**
- Roadmap for patent creation - **NPTEL**
- Digital Design with Verilog - **NPTEL**
- Simulation of Communication systems using Matlab - **NPTEL**
- Electronic Systems Design - **NPTEL**
- Introduction To Internet Of Things - **NPTEL**
- Introduction to Industry 4.0 and Industrial Internet of Things - **NPTEL**
- Fundamentals of Artificial Intelligence - **NPTEL**
- Principles of Signals and Systems - **NPTEL**
- Diploma** - Yoga for Youth Empowerment
- C Training Program - **IIT Bombay**
- JEE mains (B.E/B.Tech): Qualified in 2023 with **76.06** percentile.
- JEE mains (B.Planning): Qualified in 2023 with **88.15** percentile

Declaration

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Sanjay M